
A Biomechanical Gait Analysis in Post-Stroke Individuals and Challenges in Advanced Simulation

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ABSTRACT Stroke is one of the main causes of adult disability, which usually leads to gait impairments that restrain mobility and influences individuals' life quality [1], [2], [3]. Hemiparesis is a common post-stroke condition that causes abnormal postural and gait patterns due to muscle weakness on one side of the body [4]. With the improvement of musculoskeletal modelling and simulation platforms such as OpenSim and Moco, researchers have accessed to tools that can analyse, interpret, and predict outcomes of gait interventions [5], [6]. These tools support a range of features including inverse kinematics, joint reaction analysis, optimisation, and forward simulation that can expand our understanding of locomotor deficits in individuals who have survived stroke [7], [8]. In this project I did a biomechanical gait analysis of healthy, mild, moderate, and severe post-stroke individuals, alongside an investigation into the feasibility of advanced gait simulation using OpenSim. The goal is to quantify differences in joint reaction forces (JRFs) and muscle forces across varying stroke severities and abled bodies[9]. Motion capture and force plate data were processed using OpenSim 4.4 and custom Python scripts. Key analyses included musculoskeletal model scaling, Inverse Kinematics (IK), Inverse Dynamics (ID), Static Optimization (SO), and Joint Reaction Analysis (JRA). Results successfully demonstrated distinct JRF and muscle force patterns across the healthy, mild, moderate, and severe cases, highlighting increased asymmetry and altered force profiles with greater stroke severity. Significant technical challenges were encountered during attempts to implement advanced optimal control (OpenSim Moco) and Forward Dynamics simulations, mainly arising from persistent software installation and compatibility issues within the local computing environment. These challenges prevented the execution of intended "what-if" rehabilitation strategy simulations. Despite these computational obstacles, the early analyses provide valuable insights into the biomechanical adaptations in post-stroke gait.

INDEX TERMS Stroke, Hemiparesis, Musculoskeletal modelling, OpenSim, Kinematics, Optimisation, Joint Reaction Analysis (JRA)

I. INTRODUCTION

Stroke is a leading cause of long-term adult disability, frequently resulting in sensorimotor impairments that compromise gait, balance, and functional mobility [1], [10], [11]. Among these impairments, hemiparesis - muscle weakness or partial paralysis on one side of the body - is one of the most common and functionally limiting conditions. Hemiparesis affects over 80% of stroke survivors and is directly responsible for disrupted gait patterns, poor posture, asymmetrical loading, and increased metabolic demand during walking [2], [12]. These motor impairments often persist in spite of standard rehabilitation and significantly can affect quality of life.

Post-stroke hemiparetic gait is characterized by various biomechanical deviations, including reduced ankle dorsiflexion, prolonged stance time on the non-paretic limb, decreased push-off power in the paretic limb, and compensatory movements such as hip hiking or circumduction [1], [3]. These are commonly along with altered neuromuscular control and muscle weakness, particularly in the plantar flexors, dorsiflexors, and

quadriceps of the affected limb. The resulting asymmetries not only reduce gait efficiency but also lead to secondary problems, such as joint overuse and pain on the unaffected side.

To understand and address these issues, quantitative gait analysis methods have become essential. Technologies like three-dimensional motion capture systems, force platforms for measuring ground reaction forces (GRFs), and surface electromyography (EMG) allow for detailed biomechanical and neuromuscular assessments [13], [14]. These data provide insights into joint kinematics, kinetics, and muscle activation, forming the foundation for evidence-based rehabilitation planning.

OpenSim, an open-source musculoskeletal modeling software, enables researchers to create and simulate biomechanical models of human movement. By integrating experimental data, OpenSim can estimate joint trajectories through inverse kinematics (IK), compute joint moments using inverse dynamics (ID), calculate muscle forces through static optimization, and assess joint loading via joint reaction analysis [7], [15].

Along with OpenSim, Moco extends simulation capabilities by enabling predictive modeling of movement based on optimal control principles. It allows forward simulations in which user-defined objectives, such as minimizing joint loading or maximizing symmetry, guide the model toward physiologically realistic [5], [16].

Recent studies using predictive simulations have shown their utility in exploring the biomechanical effects of rehabilitation strategies targeting hemiparesis. Simulations have been used to model how strengthening specific muscles, adjusting joint torques, or applying assistive devices can impact gait outcomes [13], [17]. Furthermore, simulations have replicated characteristic features of hemiparetic gait, such as decreased propulsion and exaggerated swing phase asymmetries, by manipulating model parameters like muscle weakness and reflex gain [18], [19]. While promising, these tools face challenges such as model personalization, high computational cost, and limited clinical integration [20], [21].

In this project, I focused specifically on hemiparetic gait analysis. OpenSim is used to simulate gait patterns of three stroke survivors with mild, moderate, and severe hemiparesis, and two healthy subjects selected from a publicly available dataset containing full-body motion capture, and GRF data [9]. Through a series of analyses - including IK, ID, muscle force estimation, and Joint reaction forces, I aim to understand how hemiparesis affects joint mechanics and tried to explore strategies to restore functional symmetry which was not successful. Two able-bodied subjects from the same dataset serve as a control reference.

II. Review on previous works

A. Biomechanical Characterization of Post-Stroke Gait

Many studies have characterized gait deviations in individuals post-stroke, mainly by using motion capture and force plate data to quantify kinematics (joint angles, segment movements) and kinetics (joint moments, ground reaction forces). These works consistently highlight reduced walking speed, increased asymmetry, and altered joint power generation, particularly at the ankle and knee joints of the paretic limb. The studies basically consider these main classifications: stroke severity, specific gait pathologies (e.g., foot drop, knee hyperextension), and compensatory mechanisms (e.g., circumduction, hip hiking) [22], [23].

It seems more studies are tend to be more descriptive and do not delve into the underlying muscle-level contributions or the internal joint loading that drives these observable deviations. Longitudinal studies tracking changes over rehabilitation are also less common. Furthermore, traditional inverse dynamics approaches rely on experimental kinematics, which may not fully capture the dynamic interplay of muscle forces and joint stability, especially in highly compensatory gaits.[24]

B. Muscle Force Estimation in Post-Stroke Gait

To understand the muscular contributions to gait impairments, researchers have employed techniques like Static Optimization (SO) and electromyography (EMG)-driven modeling to estimate muscle forces and activations [25]. These studies often reveal altered muscle activation patterns, reduced force generation in paretic muscles, and increased co-contraction of antagonistic muscles, contributing to inefficient movement [26].

Direct validation of estimated muscle forces is challenging due to their invasiveness. Furthermore, the link between estimated muscle forces and long-term functional outcomes or specific rehabilitation interventions is not always explicitly explored. SO is an optimization technique that finds one possible solution for muscle forces and may not fully represent the complex neural control strategies employed by individuals with stroke. EMG-driven models require robust EMG data, which can be difficult to obtain and process reliably in clinical populations.[27]

C. Computational Modeling for Rehabilitation Strategy Exploration

A growing body of literature focuses on using musculoskeletal models to simulate the effects of various rehabilitation strategies. This area can be broadly clustered into:

- **Forward Dynamics Simulations:** These studies prescribe muscle excitations or joint torques to simulate movement and observe the resulting kinematics and kinetics [28]. They are useful for "what-if" scenarios, such as simulating the effect of altered muscle strength or external assistive devices.

- **Optimal Control (e.g., OpenSim Moco):** More advanced approaches, like optimal control, aim to find the most efficient or optimal way for a model to perform a task given specific objectives (e.g., minimize effort, maximize speed) and constraints (e.g., muscle strength limits) [8]. OpenSim Moco is a software toolkit designed to optimize musculoskeletal models created in OpenSim, utilizing the direct collocation method for faster and more diverse problem-solving compared to other simulation approaches. It simplifies the process by abstracting away the complexities of direct collocation, allowing researchers to focus on their scientific inquiries. Moco offers a flexible framework for defining optimal control problems through

MocoProblem and solving them with MocoSolver, which uses the CasADi library to convert continuous problems into nonlinear programs. This enables a wide range of applications in biomechanics, including motion tracking, motion prediction, and parameter optimization, and is accessible via C++, MATLAB, Python, and XML.

[6], [7]

D. Gaps and limitations

Despite the promise, a major gap lies in the direct translation of simulated optimal strategies into practical clinical interventions. The complexity of these simulations often limits their widespread use in clinical settings. There is also a need for more patient-specific models and optimization problems that truly reflect individual impairments. Furthermore, forward dynamics simulations are highly sensitive to input controls, which are difficult to prescribe accurately without experimental data or an optimization framework. Optimal control problems, while powerful, are computationally intensive and require careful tuning of objectives, constraints, and bounds to produce physiologically realistic and clinically relevant results. Technical barriers to setup and execution can also be significant, as experienced in this study.

III. Methodology

A. Data Acquisition and Subjects

In this project the data set from Tamaya Van Crielinge, A full-body motion capture gait dataset of 138 able-bodied adults across the life span and 50 stroke survivors, 2023 is used [9]. Among all the cases, three distinct individuals are chosen: two healthy subjects, table 2, one individual with mild post-stroke hemiparesis, one with moderate post-stroke hemiparesis, and one with severe case, table 1.

TABLE 1
SUBJECT CHARACTERISTICS OF STROKE SURVIVORS

ID	Age (y)	Gender (M/F)	Weight (kg)	Height (mm)	FAC	TinettiPO MA	TIS
TVC05	47	F	65	1600	2	14	13
TVC15	72	M	80	1700	5	26	16
TVC32	52	M	80	1790	3	20	14

ID: identification number, **y:** years, **M:** male, **F:** female, **kg:** kilograms
mm: millimetre, **FAC:** Functional Ambulation Category, **Tinetti POMA:** Tinetti Performance Oriented Mobility Test, **TIS:** Trunk Impairment Scale

TABLE 2
SUBJECT CHARACTERISTICS OF STROKE SURVIVORS

ID	Age (y)	Gender (M/F)	Weight (kg)	Height (mm)
SUBJ33	70	M	82	1795
SUBJ75	48	F	67	1600

ID: identification number, **y:** years, **M:** male, **F:** female, **kg:** kilograms
mm: millimetre

In the main dataset, careful selection criteria were applied to both able-bodied and post-stroke participants to ensure the validity of the results. Able-bodied individuals were excluded if they self-reported visual impairments, exhibited an antalgic gait, had abnormal lower limb mobility, or any known neurological or orthopedic disorder that could affect motor performance and balance; medical records were not collected from these volunteers.

For post-stroke individuals, went through exclusion criteria, included a Trunk Impairment Scale (TIS) score of 20 or greater (indicating normal truncal function), a Functional Ambulation Score of 2 or less (as safe gait analysis required ambulation without continuous physical support), inability to sit independently for 30 seconds on a stable surface without support or supervision, the presence of other neurological or orthopedic disorders influencing motor performance and balance, or an inability to comprehend instructions.

The Functional Ambulation Categories (FAC), Tinetti Performance-Oriented Mobility Test (POMA), and Trunk Impairment Scale (TIS) are distinct yet complementary clinical assessment tools frequently used in rehabilitation, particularly for individuals with neurological conditions like stroke. The FAC is a 6-point scale that assesses a patient's walking ability based on the level of human assistance required, regardless of assistive devices used [29].

The Tinetti POMA, also known as the Tinetti Test, is a comprehensive, task-oriented assessment that evaluates both balance and gait, providing a total score (out of 28) to indicate fall risk in older adults and various patient populations [30].

Lastly, TIS specifically measures motor impairment of the trunk by evaluating static sitting balance, dynamic sitting balance, and trunk coordination, often used to guide treatment and monitor progress in stroke rehabilitation. These tools offer valuable insights into different aspects of mobility and balance, aiding clinicians in assessing functional limitations and tailoring interventions. [29], [31].

In this project, three stroke survivors were selected and categorized into mild, moderate, and severe gait impairment groups based on their scores from the three aforementioned clinical assessments: FAC, POMA, and TIS. Subjects with high scores (FAC = 5, TIS $\approx$ 18–19, Tinetti $\approx$ 25–28) were classified as mild, indicating near-normal mobility and trunk control. Moderate cases had mid-range scores (FAC = 3–4, TIS $\approx$ 13–15, Tinetti $\approx$ 17–22), reflecting limited but independent ambulation. Severe cases were identified by low scores (FAC = 2, TIS $\approx$ 7–10, Tinetti $\leq$ 15), indicating significant balance, trunk, and walking impairments. This stratification ensured clinically meaningful comparisons in gait analysis, table 1.

All subjects performed self-selected speed walking trials, during which 3D marker trajectories and ground reaction forces (GRF) were recorded, table 3 [9].

When it comes to abled-bodied subjects, two cases are selected mainly based on their characteristic so that the male case with 82 kg body weight and 1795 mm height can be compared to the mild and moderate case while the other healthy female subject with 48 year old age, 67 kg and 1600mm height is chosen to be compared with the severe case who is similar with in characteristics.

TABLE 3
FORCE PLATE RECORDINGS FOR ABLE-BODIED AND STROKE SURVIVORS ADULTS

ID	FP 1	FP 2	FP 4
TVC05	L (BWA5)	R (BWA5)	L (BWA5)
TVC15	R (BWA3)	L (BWA2)	R (BWA3)
TVC32	R (BWA6)	L (BWA6)	R (BWA6)
SUBJ33	R (2)	L (2)	R (2)
SUBJ75	L (1)	L (1)	L (1)

L: left, R: right, FP: force plate, BWA: Basic Walk Across trials

B. Software and Tools

All musculoskeletal modeling and simulation analyses were performed using OpenSim 4.4 (Stanford University, Stanford, CA) (1). Data preprocessing, analysis, and visualization were performed using Mokka (2) and custom Python scripts (Python 3.9/3.10) leveraging key libraries including Pandas (3) for data manipulation, NumPy (4) for numerical operations, Matplotlib (5) for plotting, and Seaborn (6) for enhanced visualizations.

C. Data Preprocessing and GRF Conversion

Raw force plate data, table 3, were provided in CSV format through Mokka. A custom Python script was developed to convert these CSV files into OpenSim-compatible Ground Reaction Force (GRF) .mot files. This conversion process involved several critical steps:

- **Header Parsing:** The script was configured to dynamically identify and extract header information and column titles from the CSV file (e.g., CSV_HEADER_ROW_INDEX=9, CSV_METADATA_END_ROW_INDEX = 9). This ensured correct data parsing despite generic column names in the raw CSV.
- **Unit Conversion:** A major challenge involved inconsistent units in the raw CSV data. It was identified that force components (F_x , F_y , F_z) were in milliNewtons (mN) and moment components (M_x , M_y , M_z) were in Newton-millimeters (Nmm). For OpenSim compatibility, all forces must be in Newtons (N) and moments in Newton-meters (Nm). The Python script implemented the following conversions:
 - o Forces: $FN = FmN/1000$
 - o Moments: $MNm = MNmm/1000$
- **Center of Pressure (CoP) Calculation:** The script calculated the CoP (P_x , P_y , P_z) for each force plate

using the converted force and moment data, incorporating the global origin coordinates of each force plate. A threshold for vertical force ($F_z > 10$ N) was applied to set CoP to NaN (Not a Number) during periods of no ground contact.

- **Output Format:** The processed data, including time, forces (N), moments (Nm), and CoP (m), was written to a tab delimited .mot file ensuring compatibility with OpenSim.

D. Musculoskeletal Modeling and Kinematic Analysis

A generic musculoskeletal was scaled to each subject's anthropometry (healthy, mild, and moderate cases) using the OpenSim Scale Tool. This process involved matching model markers to experimental marker data and adjusting segment masses and inertial properties based on subject-specific measurements.

Afterwards, Inverse Kinematics (IK) was performed using the OpenSim Inverse Kinematics Tool to compute time-series joint angles (generalized coordinates) from the experimental marker data.

E. Kinetic Analysis

- **Inverse Dynamics (ID):** Joint moments were calculated using the OpenSim Inverse Dynamics Tool [1]. Inputs included the scaled model, IK results (in radians), and the processed GRF data (via an ExternalLoads.xml file, which was also generated by a custom Python script [8] to correctly link force plate data from the .mot file to model bodies).
- **Static Optimization (SO):** Muscle forces and activations were estimated using the OpenSim Static Optimization Tool. This tool takes the scaled model, IK results, and ID results as inputs, distributing the calculated joint moments among the redundant muscle actuators by minimizing the sum of squared muscle activations.
- **Joint Reaction Analysis (JRA):** Joint reaction forces and moments were computed using the OpenSim Analyze Tool with the JointReaction analysis. This analysis takes the scaled model, IK results, and Static Optimization results as inputs to determine the net forces and moments transmitted across specified joints (e.g., hip, knee, ankle), expressed in the ground coordinate system.

F. Challenges with Advanced Gait Simulation (Moco and Forward Dynamics)

Significant technical challenges were encountered during attempts to implement advanced gait simulation techniques, specifically OpenSim Moco and Forward Dynamics, within the local computing environment. These persistent issues ultimately prevented the execution of intended "what-if" rehabilitation strategy simulations.

- **OpenSim Python API for Moco:** Initial efforts focused on using the opensim.Moco Python API. Despite multiple clean reinstallations of Miniconda/Anaconda, creation of dedicated Python environments (e.g., Python 3.9), and explicit installation via `conda install -c conda-forge opensim`, the `ModuleNotFoundError: No module named 'opensim.Moco'` persisted. This indicated an unresolved underlying library dependency or compatibility issue specific to the local Windows 11 environment, preventing the Python bindings from correctly linking to the Moco C++ libraries.
- **OpenSim Moco GUI Plugin:** As an alternative, attempts were made to install the OpenSim Moco GUI Plugin for OpenSim 4.4. However, copying the plugin's .dll files into the OpenSim 4.4 bin directory led to Entry Point Not Found errors and, in some instances, corrupted the core OpenSim GUI installation, necessitating repeated reinstallation of OpenSim 4.4. This suggested a fundamental incompatibility between the plugin's build and the specific OpenSim 4.4 installation.
- **Moco XML Command Line Execution:** Even when attempting to bypass the Python API and GUI plugin by running `opensim-moco.exe` directly from the command line with manually crafted XML setup files, the tool consistently failed. Errors such as `SimTK Exception thrown at Xml.cpp:108: Error reading end tag and Encountered unrecognized Object typename...` indicated fundamental XML parsing issues or incompatibility with the `opensim-moco.exe` executable itself, despite adherence to documented XML schemas and meticulous error correction. This pointed to a deeper problem within the `opensim-moco.exe` binary's ability to interpret its own configuration files.
- **Forward Dynamics for Rehabilitation Strategies:** Due to the persistent Moco issues, an alternative approach using Forward Dynamics was pursued to simulate the effect of muscle strengthening. This involved manually modifying muscle `max_isometric_force` in the OpenSim GUI and attempting to run FD using Static Optimization results as controls. However, this also encountered challenges:
 - The `states_file` (IK results) was initially in degrees, leading to Input Storage must have values for Coordinates in meters or radians (not degrees) errors. A Python script [7] was developed to convert these to radians.
 - Despite the conversion, persistent errors indicated underlying data or parsing complexities that could not be resolved within the project timeframe.

These cumulative and persistent technical hurdles, stemming from low-level software compatibility and environment setup, prevented the successful execution of advanced optimal control and Forward Dynamics simulations.

IV. Results

After operating the mentioned operations using OpenSim tools, I saved all the results in separate files and then plotted them focusing on 3 major categories: kinematic comparison, muscle forces and joint reaction forces.

A. Kinematics Comparison

First I made a kinematic comparison, using inverse kinematics analysis results, of gait patterns among stroke patients of varying severities and against healthy subjects. The analysis focuses on key joint angles (hip flexion/extension, hip ab/adduction, knee flexion/extension, and ankle dorsi/plantar flexion) throughout the gait cycle, highlighting deviations in range of motion and timing.

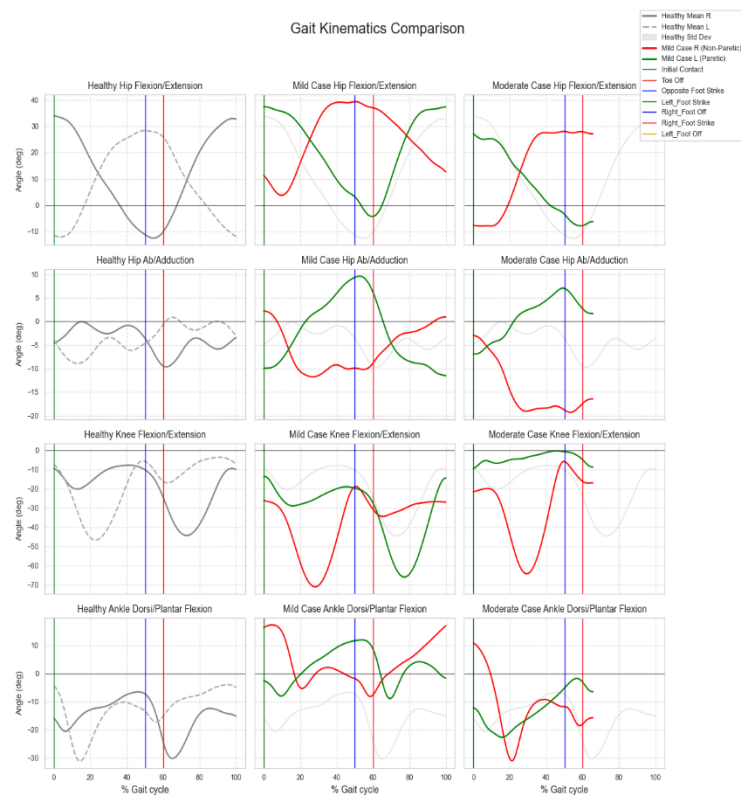


FIGURE 1. Gait Kinematics Comparison - Healthy vs. Mild and Moderate Stroke Cases : This figure illustrates the sagittal plane kinematics of the hip, knee, and ankle joints during gait for healthy individuals, mild stroke patients, and moderate stroke patients. Each row represents a joint, and columns compare the three groups. The graphs show angular displacement (degrees) over the percentage of the gait cycle, with distinct lines for healthy mean (left and right), paretic side (red), and non-paretic side (green) for stroke cases. Vertical lines indicate initial contact (blue), toe-off (green), right foot strike (orange), left foot strike (purple), right foot off (red), and left foot off (blue).

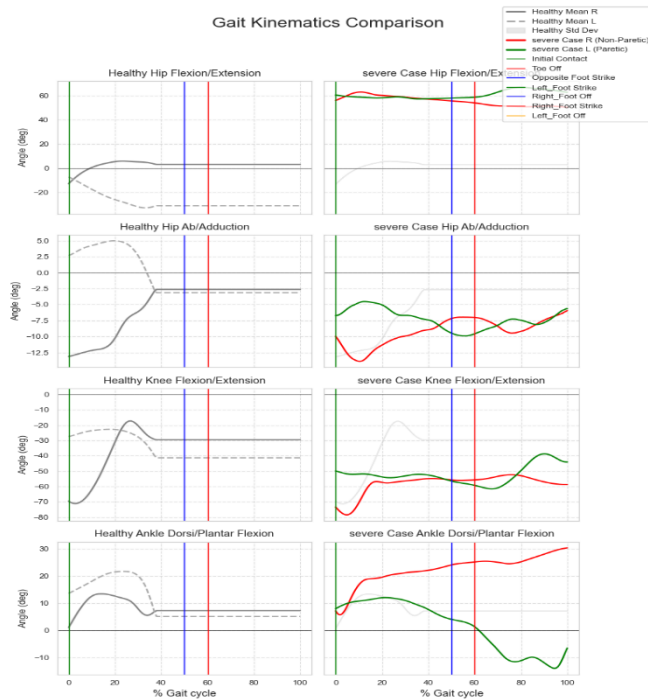


FIGURE 2. Gait Kinematics Comparison - Healthy vs. Severe Stroke Case: This figure shows the sagittal plane kinematics of the hip, knee, and ankle joints during gait, comparing healthy individuals with a severe stroke patient. Each row corresponds to a specific joint, with healthy gait patterns shown alongside the severe stroke case. The graphs plot angular displacement (degrees) against the percentage of the gait cycle, distinguishing between healthy mean (left and right), paretic side (red), and non-paretic side (green) for the severe stroke patient. Vertical lines mark key gait events: initial contact (blue), toe-off (green), right foot strike (orange), left foot strike (purple), right foot off (red), and left foot off (blue).

• Kinematic Comparison of Stroke Cases

A progressive deterioration in gait kinematics is observed with increasing stroke severity from mild to severe cases. In hip flexion/extension, mild stroke patients shows a reduced range of motion on the paretic side, characterized by decreased peak flexion during swing and limited extension during stance. This became more pronounced in moderate cases, where the paretic hip showed a significantly flatter trajectory, showing severe restriction. In severe stroke, paretic hip movement was almost negligible, often fixed in a limited range, necessitating substantial compensatory movements from the non-paretic side, which displayed an exaggerated range of motion, figure 1 and 2.

Similarly, hip ab/adduction deviations increased with severity; mild cases showed minor shifts, while moderate and severe cases presented with increased paretic adduction and compensatory non-paretic abduction.

Knee flexion/extension demonstrated a clear "stiff-knee" gait progression. Mild stroke patients showed reduced knee flexion during swing and often mild hyperextension during stance on the paretic side. Moderate cases showed more significant reductions in knee flexion, leading to a more pronounced stiff-knee appearance and greater hyperextension. In severe stroke, paretic knee flexion was considerably limited or even absent, forcing circumduction of the limb. At the same

time, the non-paretic knee showed increasingly exaggerated movements to facilitate limb advancement.

For ankle dorsi/plantar flexion, mild stroke presented with reduced dorsiflexion during swing and changed the previous contact. This foot drop worsened in moderate cases, with persistent plantarflexion. Severe stroke patients displayed severe equinus deformity, characterized by fixed plantarflexion and a complete absence of dorsiflexion for toe clearance, further needed compensatory strategies from the non-paretic limb.

• Comparison of Healthy Male Subject with Mild and Moderate Stroke Cases

When comparing healthy male gait kinematics to mild stroke patients, distinct differences emerged. Healthy individuals clearly showed smooth, symmetrical, and full ranges of motion across all joints. In mild stroke, the paretic limb showed a noticeable reduction in hip flexion/extension range, with a less pronounced peak flexion and extension compared to healthy controls, figure 1.

Hip ab/adduction in mild cases displayed slight shifts towards adduction on the paretic side, differing from the more centered and minimal movement in healthy gait.

Knee flexion/extension in mild stroke was characterized by reduced knee flexion during the swing phase and a tendency towards hyperextension during standing, contrasting with the healthy double-knee flexion pattern.

Ankle dorsi/plantar flexion in mild stroke showed a clear "foot drop" during swing, with insufficient dorsiflexion for toe clearance, and often a plantarflexed initial contact, unlike the controlled dorsiflexion and heel-strike observed in healthy gait.

In moderate stroke case, these deviations from healthy male kinematics were even more noticeable. The paretic hip flexion/extension curve was clearly flattened, stating severe limitation in movement.

Hip ab/adduction showed more pronounced paretic adduction and compensatory non-paretic abduction. The "stiff-knee" gait was highly evident, with severe limitations in knee flexion during swing and persistent hyperextension during stance, leading to a much straighter limb trajectory compared to healthy controls.

Ankle dorsi/plantar flexion in moderate stroke demonstrated severe foot drop and often a fixed plantarflexion, making toe-off and initial contact highly abnormal relative to the healthy pattern. The non-paretic limb in both mild and moderate cases often showed compensatory adjustments, but these were more pronounced in moderate stroke, reflecting the greater impairment of the paretic side.

- **Comparison of Healthy Female Subject with Severe Stroke Case**

The comparison between healthy female gait and severe stroke gait depicted major kinematic impairments. Healthy female gait exhibited typical, fluid, and symmetrical joint movements. In stark contrast, the severe stroke patient's paretic limb displayed extremely limited and often abnormal motion, figure 2.

Hip flexion/extension on the paretic side was severely restricted, often appearing as a nearly flat line with minimal angular displacement, indicating a near-fixed hip joint. This necessitated substantial hip abduction on the non-paretic side to clear the paretic limb during swing, a marked deviation from the minimal hip ab/adduction in healthy gait.

The paretic knee flexion/extension in severe stroke was profoundly impaired, with an almost complete absence of knee flexion during swing, resulting in a severe "stiff-knee" gait. This often led to compensatory circumduction of the entire limb to achieve foot clearance, a movement entirely absent in healthy gait.

Furthermore, ankle dorsi/plantar flexion on the paretic side was characterized by severe equinus deformity, where the ankle remained in a plantarflexed position throughout the gait cycle, preventing proper heel strike and toe-off. This severe foot drop and lack of ankle dorsiflexion significantly impacted ground clearance and stability, standing in sharp contrast to the controlled and dynamic ankle movements observed in healthy female gait. The non-paretic limb in severe stroke exhibited highly exaggerated movements across all joints to compensate for the profound limitations of the paretic side, demonstrating the significant adaptive strategies employed to maintain ambulation.

B. Muscle Forces

After kinematics comparison, I analyzed the findings from the muscle force analysis, derived through static optimization, comparing various stroke severities and healthy subjects. The interpretation focuses on the magnitude and temporal patterns of muscle activation to understand the compensatory strategies and limitations in force generation during gait, figure 3.

- **Muscle Force Comparison Across Stroke Cases**

The analysis of muscle forces revealed a clear progression of impairment and compensatory strategies with increasing stroke severity. In mild stroke cases, the paretic limb generally exhibited reduced peak muscle forces, particularly in the ankle dorsiflexors (e.g., Tibialis Anterior) during swing, consistent with the observed foot drop. There was also a tendency for decreased extensor muscle activity (e.g., Quadriceps, Gastrocnemius) during the stance phase, leading to less efficient propulsion. Compensatory increases in force were sometimes noted in the non-paretic limb or in proximal muscles (e.g., hip flexors) to aid limb advancement.

For moderate stroke cases, the reductions in paretic muscle forces were more pronounced. Key propulsive muscles like the Gastrocnemius and Soleus showed significantly diminished activity, impacting push-off. Similarly, quadriceps activity was often reduced, contributing to knee instability or hyperextension. The non-paretic limb demonstrated greater compensatory efforts, with elevated force magnitudes in its hip and knee extensors to bear more weight and generate forward momentum. The overall muscle force patterns became less rhythmic and more irregular, reflecting increased effort and decreased efficiency.

In severe stroke cases, the paretic limb displayed a profound inability to generate meaningful muscle forces in many key gait muscles. The force profiles for paretic ankle and knee musculature were often minimal or absent, consistent with the severe equinus deformity and stiff-knee gait observed kinematically. This necessitated extreme compensatory force generation from the non-paretic limb and trunk muscles to achieve any form of ambulation. The muscle force patterns in severe stroke were highly disorganized and asynchronous, indicative of severe motor control deficits and reliance on gross, inefficient movements.

- **Comparison of Healthy Male Subject with Mild and Moderate Stroke Cases**

Comparing the healthy male's muscle force patterns to those of mild stroke patients highlighted specific deficits. The healthy male exhibited robust, symmetrical, and well-timed peaks in muscle activity, reflecting efficient force generation for propulsion, support, and limb advancement. For instance, clear peaks were observed in the quadriceps during early stance for knee stability, in the hamstrings during late swing for deceleration, and in the gastrocnemius/soleus complex during late stance for push-off.

In mild stroke, the paretic limb showed reduced peak forces and altered timing compared to the healthy male. Specifically, the paretic Tibialis Anterior often had lower and less sustained activity during swing, contributing to foot drop. Paretic plantarflexors and quadriceps also showed reduced peak forces, impairing push-off and stance phase stability. The non-paretic limb, while generally stronger than the paretic side, might display slightly increased forces in some phases as a compensatory mechanism, though less dramatically than in more severe cases.

Moderate stroke patients exhibited even greater deviations from the healthy male's muscle force profiles. The paretic limb's muscle forces were significantly attenuated across most major gait muscles, with particularly low or absent peaks in the ankle plantarflexors and knee extensors. This indicated a substantial loss of propulsive power and weight-bearing capacity. The non-paretic limb in moderate stroke patients showed more pronounced compensatory increases in muscle forces, particularly in the hip and knee extensors, to propel the body forward and maintain stability, reflecting a greater reliance on the unaffected side for ambulation.

Comparison of Healthy Female with Severe Stroke Case

The comparison between healthy female muscle forces and those of a severe stroke patient revealed a stark contrast in motor function. The healthy female demonstrated well-coordinated, rhythmic, and appropriately scaled muscle activation patterns across all phases of gait, ensuring smooth and efficient movement.

In the severe stroke case, the paretic limb's muscle force generation was profoundly compromised. Many paretic muscles, especially those critical for ankle dorsiflexion (Tibialis Anterior) and plantarflexion (Gastrocnemius, Soleus), showed minimal to no discernible force production throughout the gait cycle. This absence of effective muscle activation directly corresponded to the observed severe foot drop and equinus deformity. Similarly, paretic knee and hip musculature exhibited severely diminished or absent force peaks, indicating a near-complete inability to contribute to limb support or advancement. Consequently, the non-paretic limb of the severe stroke patient displayed dramatically increased and often prolonged muscle activation, particularly in the hip and knee extensors, to compensate for the paretic limb's inability to generate force. This highly asymmetric and inefficient muscle force distribution underscored the severe motor impairment and the significant compensatory burden placed on the unaffected side to achieve even limited ambulation.

C. Joint Reaction Forces

Finally, I plotted the joint reaction force analysis results, providing insights into the internal loads experienced by the hip, knee, and ankle joints during gait across different stroke severities and in comparison, to healthy controls, figure 4 and 5.

• Joint Reaction Force Comparison Across Stroke Cases

The analysis of joint reaction forces (JRFs) revealed a progressive shift in load distribution and magnitude with increasing stroke severity. In mild stroke cases, the paretic limb generally exhibited slightly reduced peak JRFs, particularly in the vertical (F_z) direction, at the knee and ankle joints during the stance phase. This reduction suggests a diminished ability of the paretic limb to bear weight effectively. Compensatory increases in JRFs were sometimes observed on the non-paretic side, indicating a greater reliance on the unaffected limb for support and propulsion. The temporal patterns of JRFs on the paretic side also showed minor alterations, with less distinct peaks compared to healthy gait.

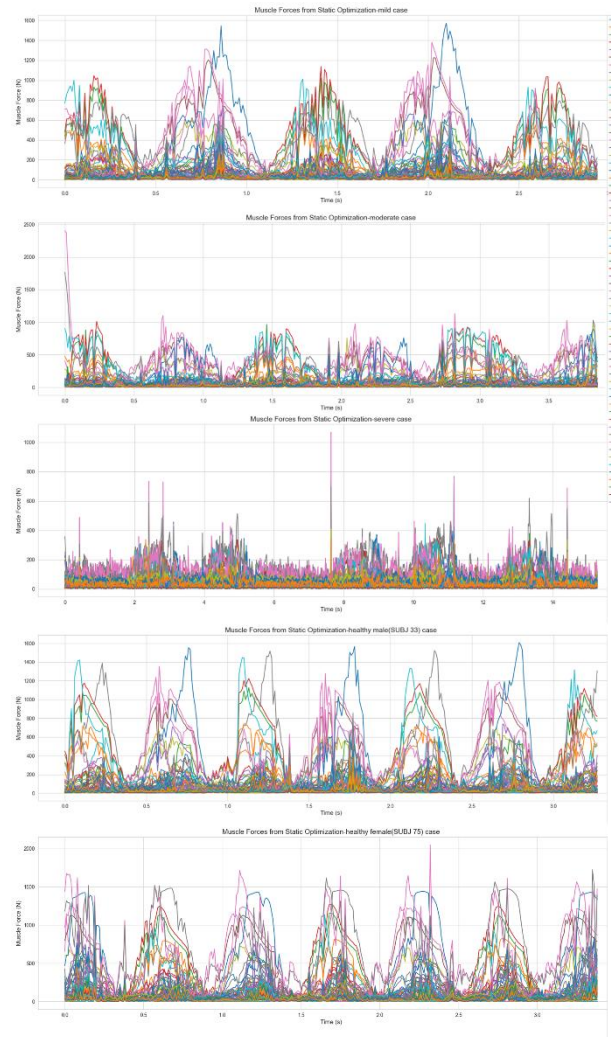


FIGURE 3. Muscle Forces from Static Optimization - All Cases: This figure presents the muscle forces derived from static optimization across various gait conditions. It includes data for mild, moderate, and severe stroke cases, as well as a healthy male and a healthy female. Each subplot displays the force profiles of multiple muscles over time, providing a comprehensive overview of muscle activation patterns and magnitudes during gait for different subject groups.

For moderate stroke cases, the reductions in paretic limb JRFs were more pronounced. Peak vertical JRFs at the paretic knee and ankle were significantly lower and often broader, indicating a prolonged and less efficient weight-bearing phase. The anterior-posterior (F_x) and medial-lateral (F_y) forces also showed greater asymmetry and altered magnitudes, reflecting impaired balance and control. The non-paretic limb in moderate stroke patients consistently demonstrated higher peak JRFs, particularly at the hip and knee, as it assumed a greater load-bearing role to compensate for the paretic limb's weakness and instability.

In severe stroke cases, the JRF patterns on the paretic limb were severely attenuated and highly irregular. The vertical JRFs at the paretic ankle and knee were often minimal, reflecting a profound inability to bear weight or generate effective ground reaction forces. The force profiles for all directions (Fx, Fy, Fz) were highly disorganized, lacking the characteristic rhythmic peaks seen in healthy gait. This indicated severe instability and a minimal contribution of the paretic limb to body support. Consequently, the non-paretic limb in severe stroke patients bore a disproportionately high load, with significantly elevated and sustained JRFs across all joints, particularly the hip and knee, as it became the primary means of support and propulsion.

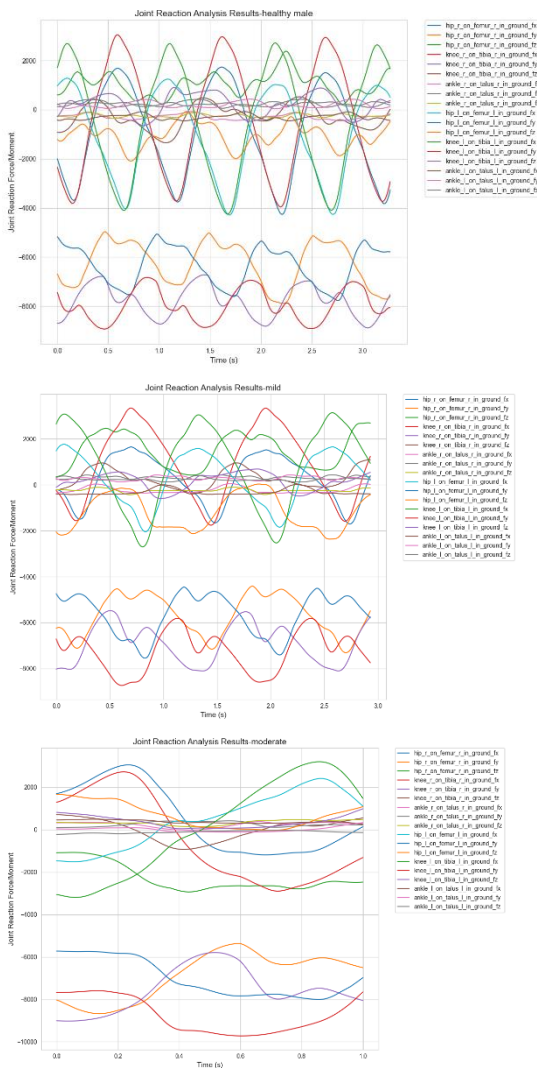


FIGURE 4. Joint Reaction Analysis Results for the healthy male subject, mild and moderate case: This figure presents the joint reaction forces and moments during each case's gait. The plots show the forces in the X, Y, and Z directions at the hip, knee, and ankle joints for both the right and left limbs, over time.

• Comparison of Healthy Male with Mild and Moderate Stroke Cases

When comparing the joint reaction forces of a healthy male to those of mild stroke patients, clear differences in load distribution were observed. The healthy male exhibited symmetrical and well-defined JRF profiles with characteristic double peaks in vertical forces at the hip, knee, and ankle during the stance phase, indicating efficient load transfer and shock absorption. The anterior-posterior and medial-lateral forces also showed balanced and rhythmic patterns.

In mild stroke, the paretic limb's JRFs, particularly the vertical components, showed slightly reduced magnitudes and less distinct peaks compared to the healthy male. This suggested a subtle decrease in weight acceptance and push-off capabilities. The non-paretic limb's JRFs often showed a slight increase in magnitude, indicating a compensatory shift in weight bearing, though the overall symmetry was largely maintained compared to more severe cases.

For moderate stroke, the deviations from the healthy male's JRF patterns were significantly more pronounced. The paretic limb's vertical JRFs were substantially lower and often lacked clear peaks, indicating a severe reduction in weight-bearing capacity and inefficient load transfer. The anterior-posterior and medial-lateral forces on the paretic side also showed greater irregularities and asymmetries, reflecting impaired balance and control. Conversely, the non-paretic limb in moderate stroke patients consistently displayed higher peak JRFs across all joints and directions compared to the healthy male, demonstrating a significant compensatory burden to maintain stability and propulsion.

Comparison of Healthy Female with Severe Stroke Case

The comparison between healthy female joint reaction forces and those of a severe stroke patient revealed profound differences in joint loading and gait mechanics. The healthy female displayed symmetrical, rhythmic, and appropriately scaled JRFs across all joints, indicative of efficient and balanced load distribution during normal ambulation.

In stark contrast, the severe stroke patient's paretic limb exhibited extremely low and highly irregular JRFs. The vertical forces at the paretic ankle and knee were often negligible, signifying a near-complete inability of the limb to support body weight. The anterior-posterior and medial-lateral forces on the paretic side were also minimal and disorganized, reflecting a severe lack of control and stability. This profound impairment meant that the non-paretic limb bore a disproportionately high load. The non-paretic hip, knee, and ankle joints of the severe stroke patient showed significantly elevated and often prolonged JRFs in all directions compared to the healthy female. This extreme asymmetry in joint loading underscores the severe functional limitations of the paretic limb and the substantial compensatory demands placed on the unaffected side to facilitate any form of locomotion.

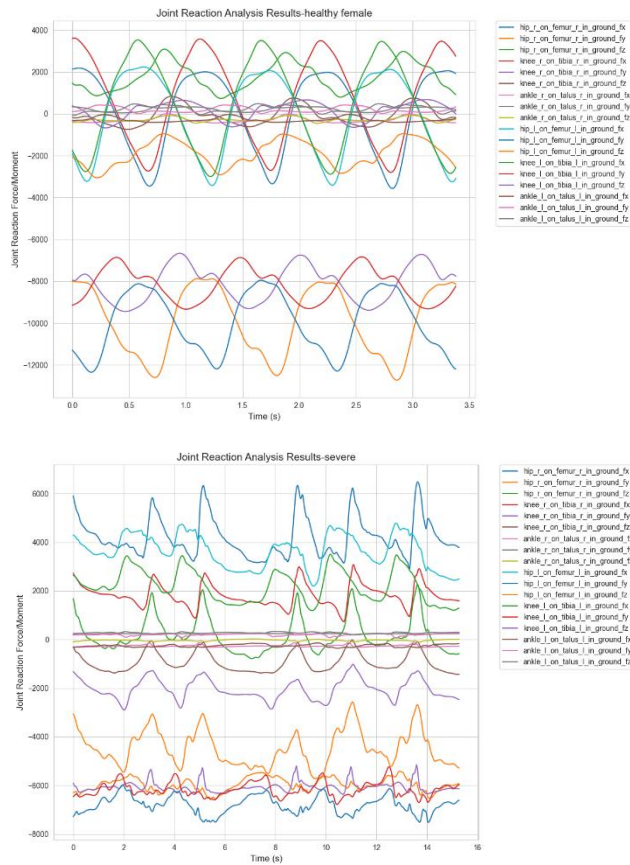


FIGURE 5. Joint Reaction Analysis Results for the healthy female subject and severe case: This figure presents the joint reaction forces and moments during each case's gait. The plots show the forces in the X, Y, and Z directions at the hip, knee, and ankle joints for both the right and left limbs, over time.

V. CONCLUSION

In conclusion, this comprehensive biomechanical analysis, integrating gait kinematics, muscle force estimations, and joint reaction force calculations, provides compelling evidence of a clear and progressive deterioration in locomotor function directly correlated with increasing stroke severity. The kinematic data consistently revealed significant deviations from healthy gait patterns, including a marked reduction in the range of motion at the hip, knee, and ankle joints on the paretic side. Crucially, altered temporal sequencing of joint movements and the emergence of compensatory strategies such as "stiff-knee" gait, pronounced foot drop, and circumduction were observed, becoming increasingly severe and functionally limiting in moderate and severe stroke cases. These kinematic impairments directly hinder efficient limb advancement and weight transfer during ambulation.

The muscle force analysis, derived through static optimization, provided a critical mechanistic link to the observed kinematic deficits. It demonstrated a profound and progressive reduction in the ability of paretic limb musculature to generate effective forces, particularly in key propulsive and stabilizing muscle groups (e.g., ankle plantar flexors, knee extensors). This diminished force production

necessitated a significant and increasing reliance on compensatory muscle activation from the non-paretic limb and trunk. The non-paretic side often exhibited exaggerated force magnitudes and prolonged activity to compensate for the paretic limb's inability to contribute adequately to propulsion and support, leading to an overall inefficient and asymmetrical gait pattern.

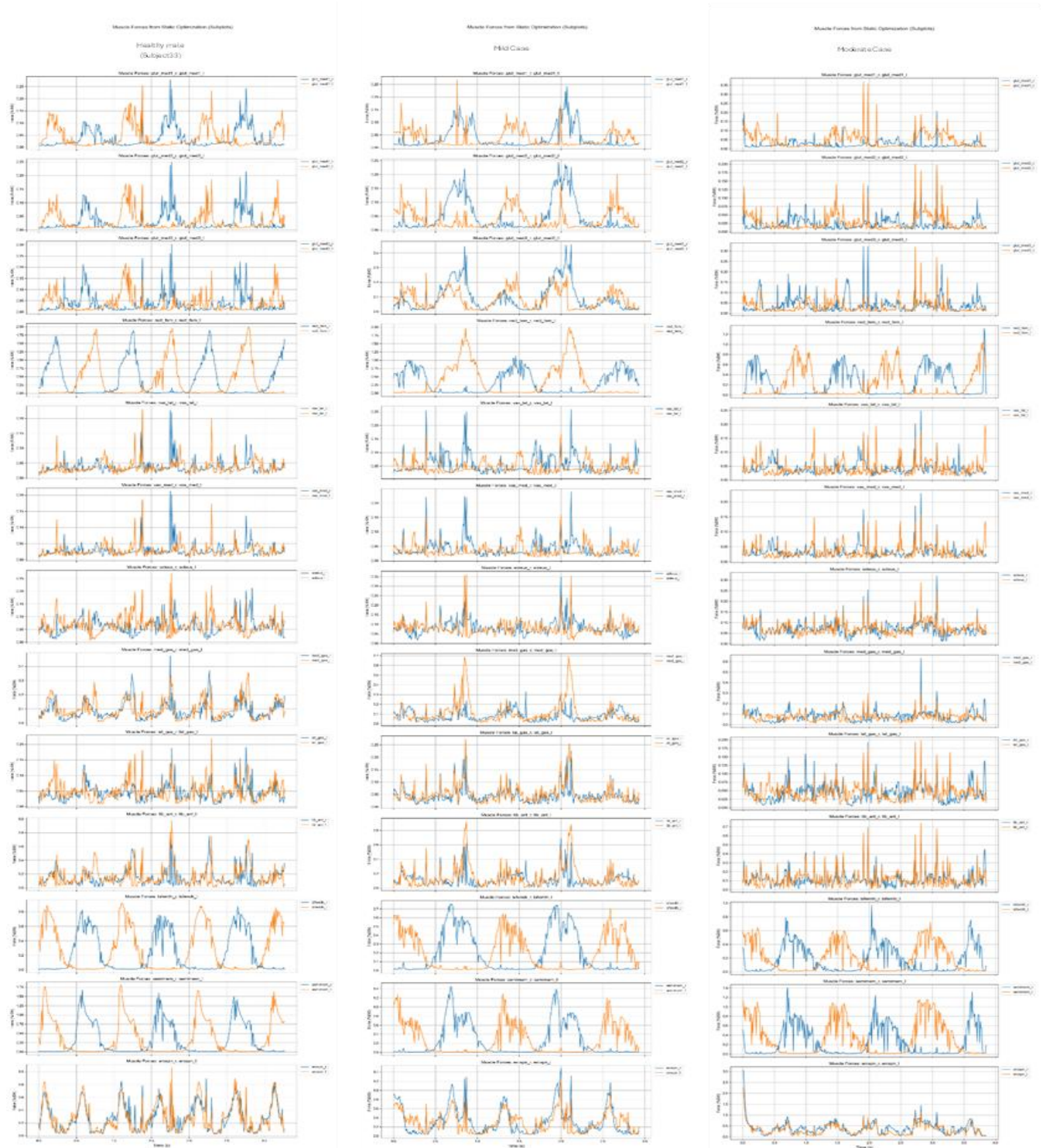
Furthermore, the joint reaction force (JRF) data elucidated the altered load distribution within the lower limb joints. A consistent finding was the reduced loading on the paretic joints, particularly in the vertical direction, indicating a diminished capacity for weight acceptance and bearing. Conversely, the non-paretic limb consistently experienced substantially increased JRFs across its hip, knee, and ankle joints. This asymmetrical loading pattern suggests a heightened risk of overuse injuries and degenerative changes in the non-paretic limb over time, in addition to contributing to overall gait instability.

Collectively, these findings underscore the profound and multifaceted impact of stroke on the entire gait cycle, extending beyond primary motor deficits to influence compensatory strategies and joint loading throughout the lower extremities. This detailed understanding of biomechanical alterations highlights the critical need for targeted and individualized rehabilitation strategies. Such interventions should not only focus on restoring strength and motor control in the paretic limb but also address the maladaptive compensatory patterns developed by the non-paretic limb. The ultimate goal of these strategies should be to improve overall walking efficiency, enhance stability, and mitigate the long-term consequences of asymmetric joint loading, thereby improving the quality of life for stroke survivors.

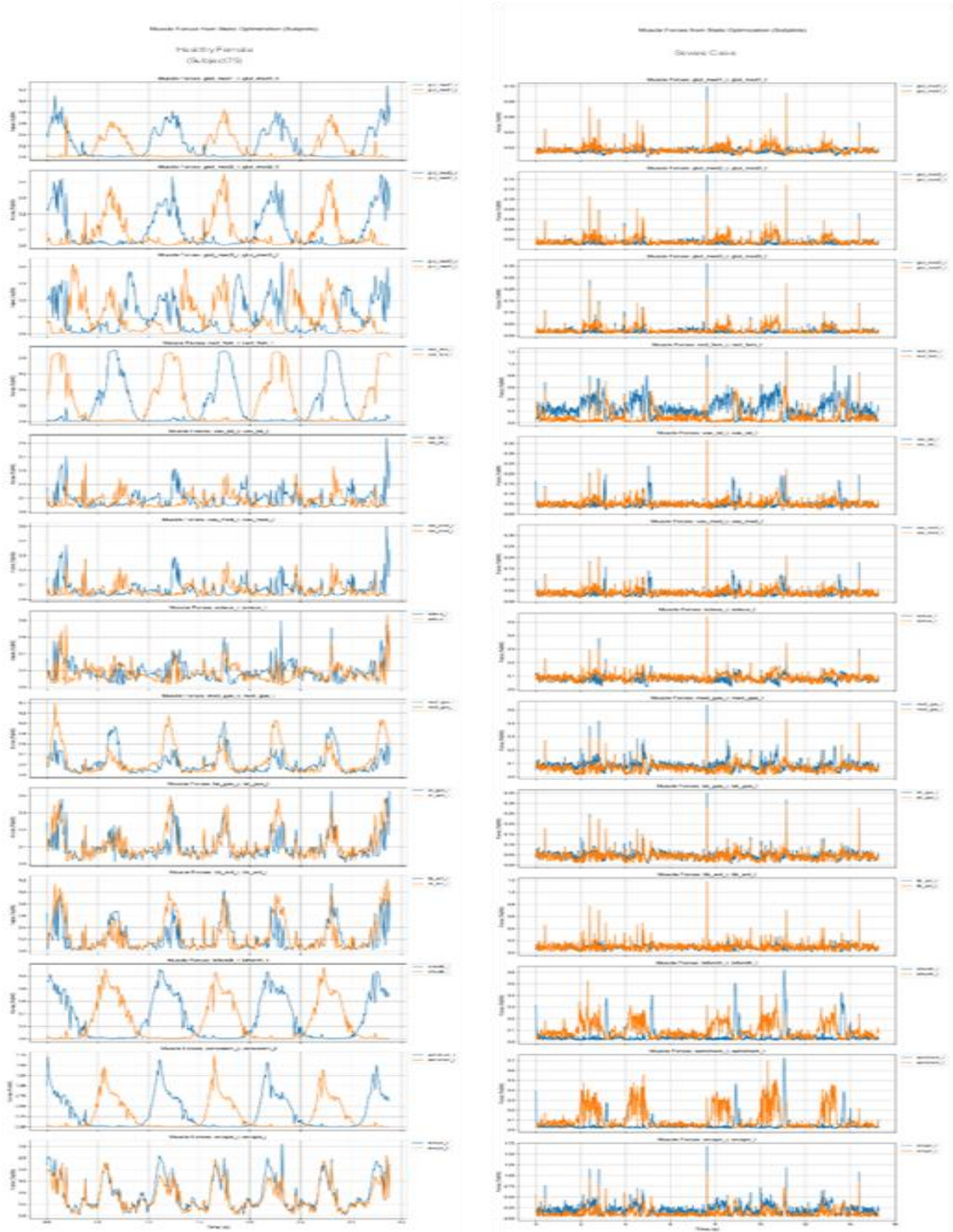
Future work will aim to leverage these insights to develop more effective rehabilitation interventions. While initial attempts at simulating rehabilitation strategies using tools like MoCo or Forward Dynamics (FD) have proven challenging, further refinement of musculoskeletal models and optimization techniques is warranted. Overcoming these computational hurdles will be crucial for predicting the effects of various interventions and designing personalized rehabilitation protocols that can optimize gait outcomes and minimize long-term complications for stroke patients.

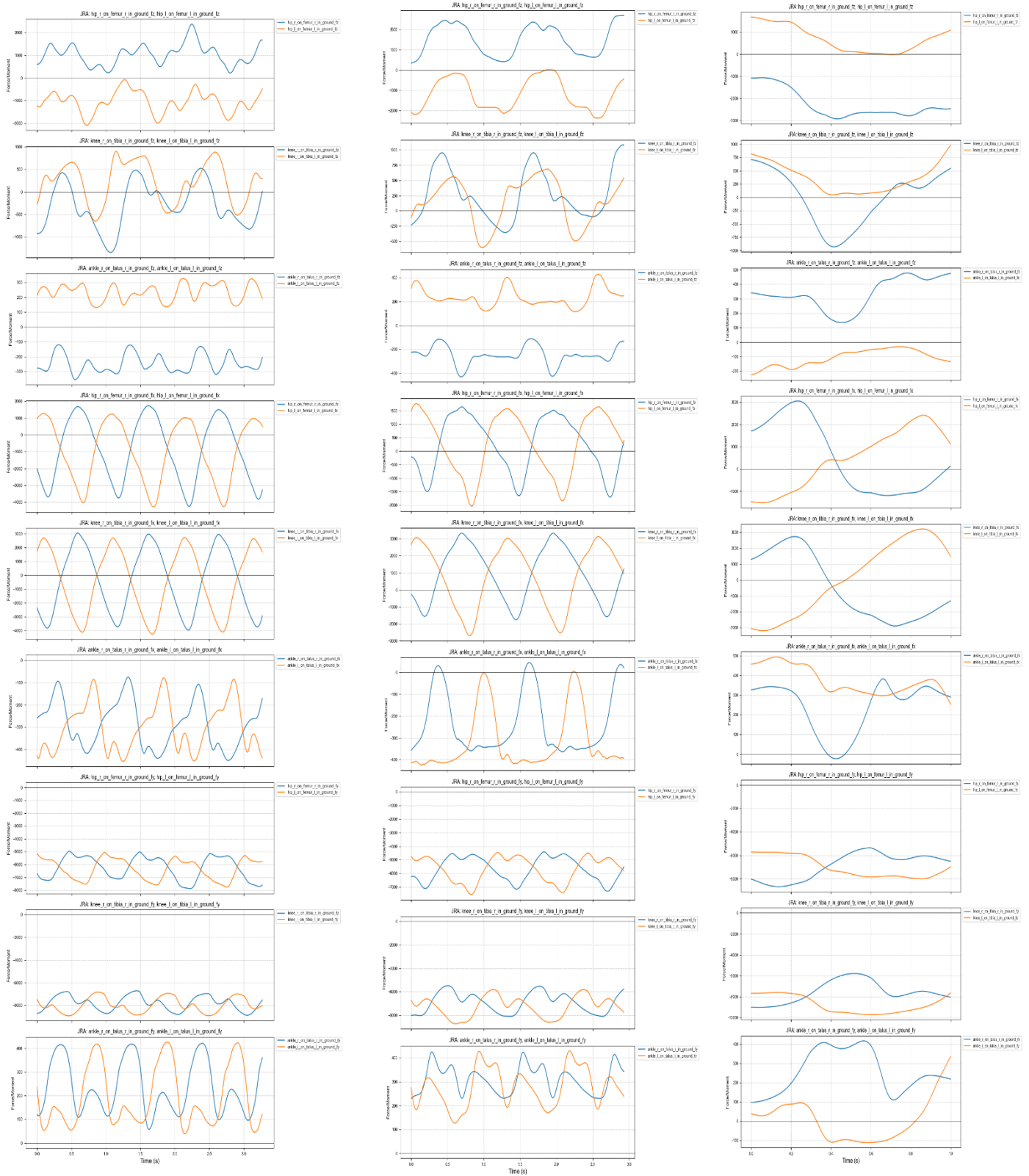
APPENDIX A Subplots

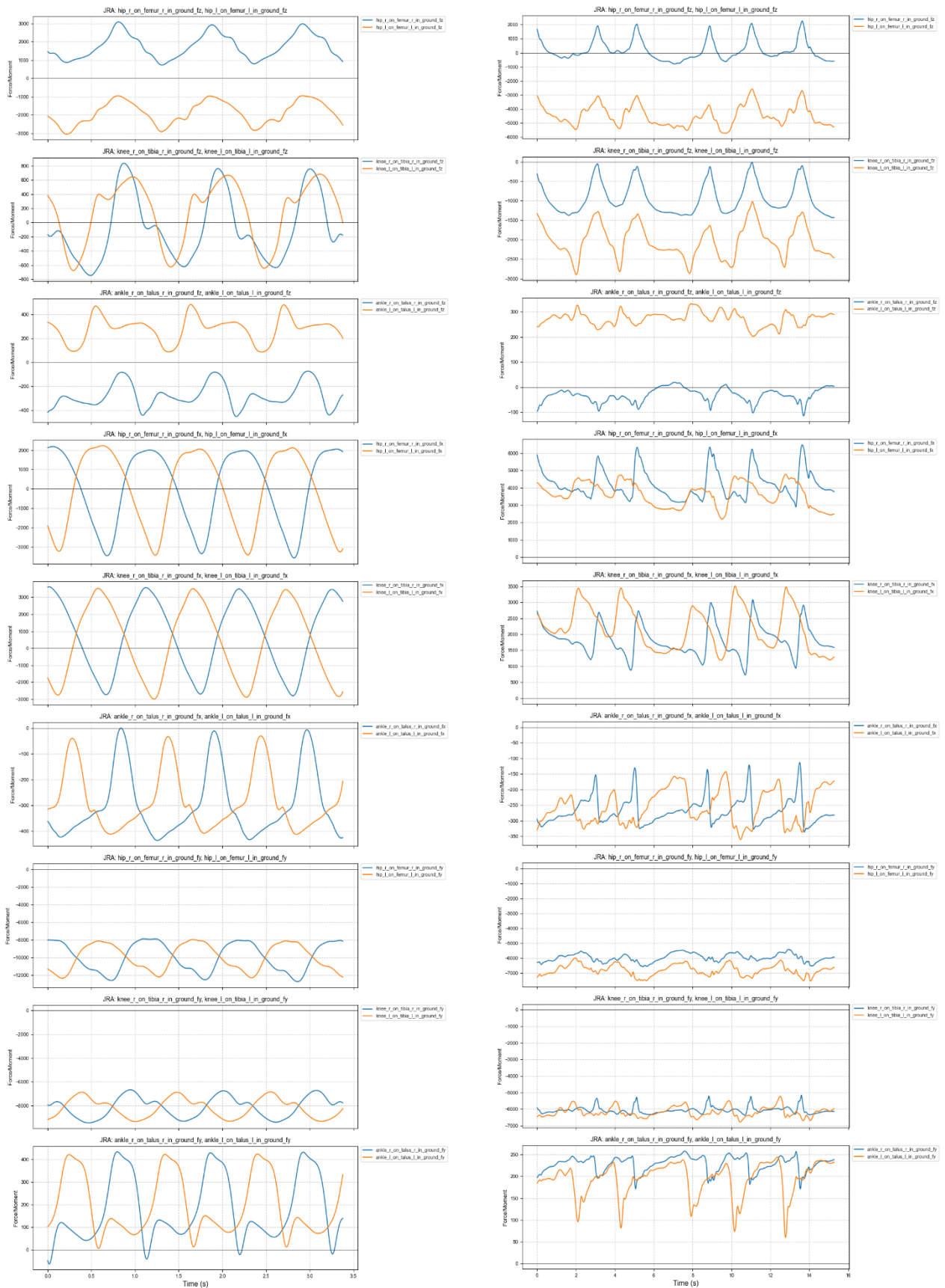
Plot 1: Muscle Forces from Static Optimization - Healthy Male vs. Mild and Moderate Stroke Cases: This figure compares the muscle forces obtained from static optimization for a healthy male individual against those of mild and moderate stroke patients. The plots illustrate the force profiles of various muscles over time, highlighting differences in muscle activation strategies and magnitudes between healthy gait and the two levels of stroke severity. Each column represents a different subject group (healthy male, mild stroke, moderate stroke), allowing for direct comparison of muscle force contributions.



Plot 2: Muscle Forces from Static Optimization - Healthy Female vs. Severe Stroke Case: This figure displays the muscle forces derived from static optimization, comparing a healthy female individual with a severe stroke patient. The plots show the force profiles of different muscles over time, emphasizing the significant alterations in muscle activation patterns and magnitudes in severe stroke compared to healthy gait. The left column represents the healthy female, and the right column represents the severe stroke case, facilitating a clear visual comparison of muscle force contributions.



Plot 3: Joint Reaction Forces: Healthy male subject, Milde and Moderate cases.

Plot 4: Joint Reaction Forces: Healthy Female and severe case.

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